



MDR Transaction & Cost Analysis

Data period: Apr-Sep 2026

Recruiter-facing portfolio case study | SQL Server + Python + Power BI + DAX



Total Transactions
1,000



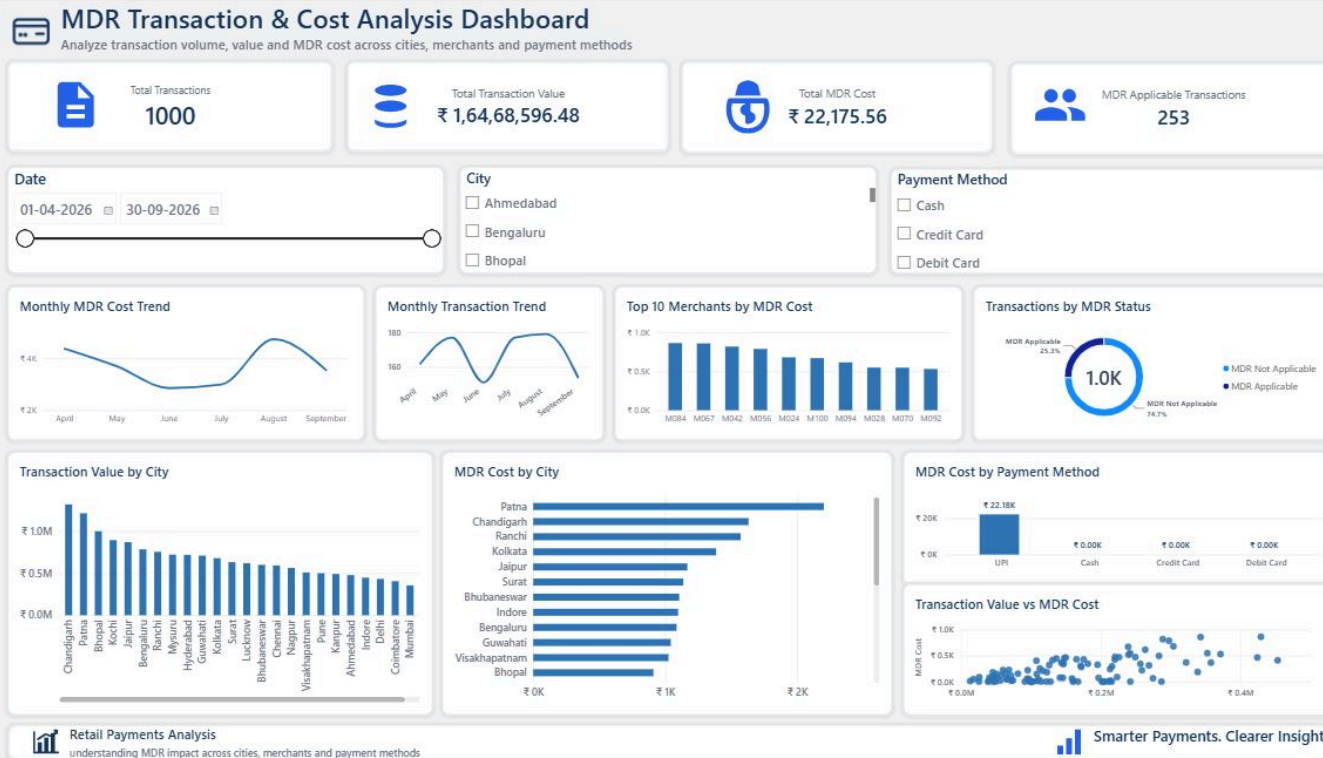
Total Transaction Value
₹ 1,64,68,596.48



Total MDR Cost
₹ 22,175.56



MDR Applicable Transactions
253



Interactive dashboard: date, city and payment-method filters with merchant, city, trend and MDR views

Built as an end-to-end data analyst portfolio project

From raw transactions to a decision-ready model

End-to-end workflow

The project combines data preparation, SQL validation, dimensional modeling and DAX into one reproducible analytics workflow.

Project approach

1 Prepare & validate the transaction data
Checked transaction amounts, merchant/payment/city/date keys and MDR eligibility fields; verified totals and row counts before reporting.

2 Apply MDR business logic
Separated MDR-applicable and MDR-exempt transactions and calculated MDR cost at transaction level.

3 Model the data in Power BI
Built a star schema with six dimensions connected directly to the fact view using one-to-many, single-direction relationships.

4 Build & validate measures
Created eight DAX measures, then cross-checked Power BI totals against SQL results before finalizing the dashboard.

SQL Server

Python

Power BI

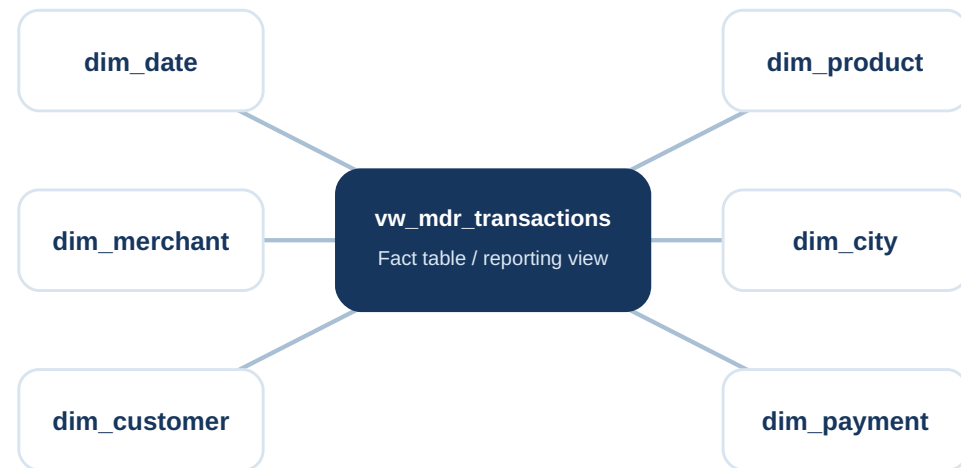
DAX

Star Schema

Data Validation

Validation checkpoint: Power BI matched the SQL outputs - 1,000 transactions, ₹1,64,68,596.48 total transaction value, 253 MDR-applicable transactions and ₹22,175.56 total MDR cost.

Star schema



Core DAX measures

Total Transactions

Total MDR Cost

MDR Applicable Transaction Value

MDR Exempt Transactions

Total Transaction Value

MDR Applicable Transactions

MDR Rate

MDR Exempt Transaction Value

What the analysis revealed

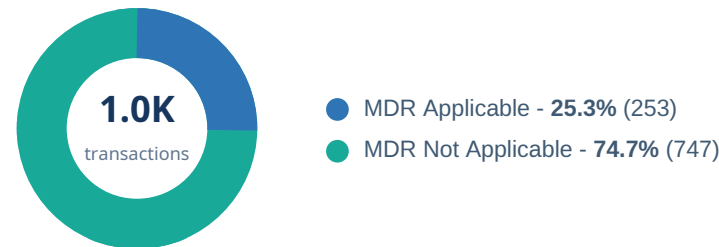
Key findings

The dashboard turns the transaction-level data into a concise set of operational and cost-management insights.

1. Most transactions are not MDR-applicable

74.7% not applicable

Only 253 of 1,000 transactions are MDR-applicable; 74.7% are not. The remaining share is not MDR-applicable after eligibility and exemption rules.



2. UPI drives recorded MDR cost

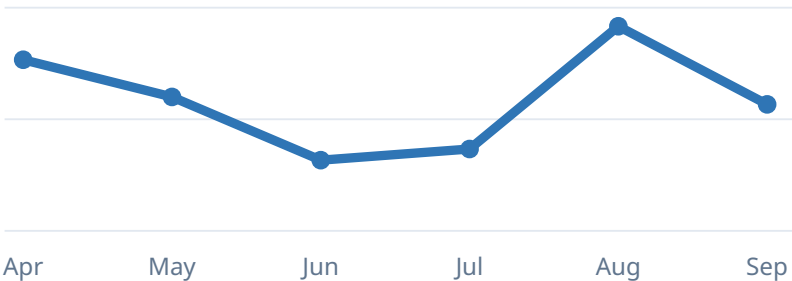
100% of recorded MDR cost

The payment-method view shows ₹22,175.56 of MDR cost on UPI, while Cash, Credit Card and Debit Card show zero recorded MDR cost.



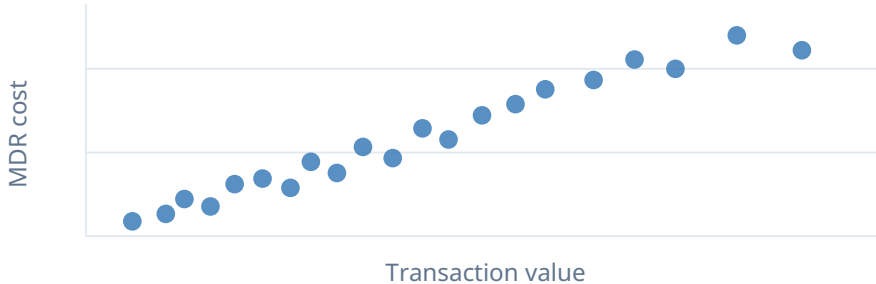
3. MDR cost varies materially by month

The monthly cost curve declines from April into June, recovers in July, peaks in August, then softens in September - useful for spotting changes in transaction mix or MDR applicability.



4. Cost concentration can be investigated by merchant and city

The Top 10 merchant and city views reveal where MDR cost is concentrated. The scatter plot also shows that higher transaction value generally accompanies higher MDR cost, while still leaving visible merchant-level variation.



Business value & portfolio takeaways

Outcome

The project demonstrates not only dashboarding, but the full chain from business logic to validated decision support.

Recommended business actions

Prioritize UPI cost monitoring

Because all recorded MDR cost in this dataset comes from UPI, payment-method monitoring should make UPI the first focus for cost investigation.

Watch high-cost merchants & cities

Use the Top 10 merchant and city views to set review thresholds and investigate unusual increases rather than treating MDR cost as evenly distributed.

Track the MDR applicability mix

With 74.7% of transactions not applicable, monitor how eligibility and exemption rules affect the overall cost base and how that share changes over time.

Use slicers for root-cause analysis

Date, city and payment filters let business users isolate a spike quickly and compare the same KPI set without requesting a separate report.

Project outcome: a validated, interactive reporting model that connects transaction behavior to MDR cost and makes concentration, trends and applicability patterns visible at a glance.

Skills demonstrated

SQL querying & validation

Python data preparation

Dimensional modeling

Power BI report design

DAX measures

KPI definition

Business analysis

Data storytelling

Deliverables



Interactive Power BI dashboard

KPI cards, slicers, trend, merchant, city, payment-method, MDR-status and scatter analysis.



Validated star-schema model

Six dimensions connected to a central transaction fact view with clean one-to-many relationships.



Reusable DAX measure layer

Eight business measures organized in a dedicated measure table for maintainability.



Recruiter-ready case study

A concise story focused on the problem, method, findings and business value rather than interview preparation.